CME and Solar Flare Transient F_U Perturbation Dynamics

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PAPER_1081: CME and Solar Flare Transient F_U Perturbation Dynamics

Abstract

Coronal mass ejections (CMEs) inject $\sim 10^{32}$ J over hours, transiently boosting solar wind density by 10--$100\times$ and magnetic field by 10--$1000\times$. We derive the transient F_U perturbation through three channels: (i) dipole perturbation ΔU_{g1} from flare-enhanced B_s, (ii) ram pressure perturbation ΔU_{g2} from CME density/velocity, and (iii) magnetic string flux boost ΔU_m. The perturbed field $F_U^{\text{pert}} = F_U^{\text{quiet}} + \Delta U_{g1} + \Delta U_{g2} + \Delta U_m$ shows amplification factors $\sim 10^5$ during Carrington-class events.

§1 Dipole Perturbation

$$\Delta U_{g1} = \Delta\mu_s \cdot \nabla\left(\frac{M_s}{r}\right) \cdot e^{-\alpha_f \Delta t}$$

where $\Delta\mu_s = (B_{\text{flare}} - B_{\text{quiet}}) \cdot R_s^3$ is the excess dipole moment from the flare-enhanced surface field, and $\alpha_f \sim 0.01$ s^{-1} governs the fast exponential recovery.

Typical values:
- $B_{\text{flare}} \approx 0.4$ T (4000 Gauss), $B_{\text{quiet}} \approx 10^{-4}$ T
- $\Delta\mu_s \approx 0.4 \cdot (6.96 \times 10^8)^3 \approx 1.35 \times 10^{26}$ A$\cdot$m^2

§2 CME Ram Pressure Perturbation

$$\Delta U_{g2} = \frac{\rho_{\text{CME}} \cdot v_{\text{CME}}^2 \cdot A_{\text{CME}}}{4\pi d^2}$$

CME Class	E (J)	v (m/s)	ρ (kg/m^3)	ΔU_{g2} (m/s^2)
Carrington	10^{33}	2.5×10^6	10^{-18}	$\sim 10^{-7}$
Moderate	10^{31}	5×10^5	5×10^{-20}	$\sim 10^{-9}$
Micro-flare	10^{28}	3×10^5	10^{-20}	$\sim 10^{-11}$

§3 Magnetic String Flux Boost

$$\Delta U_m = N_{\text{str}} \cdot \frac{\Delta\mu_j}{r_j} \cdot \left(1 - e^{-\gamma_f \Delta t}\right)$$

where $N_{\text{str}} \sim 10^9$ strings and $\gamma_f \sim 0.005$ s^{-1}.

§4 Total Perturbation

$$F_U^{\text{pert}} = F_U^{\text{quiet}} + \Delta U_{g1} + \Delta U_{g2} + \Delta U_m$$

At $\Delta t = 0$ (eruption onset):

$$\text{Amplification} = \frac{F_U^{\text{pert}}}{F_U^{\text{quiet}}} \approx 1.58 \times 10^5$$

The dominant contribution is ΔU_{g1} from the flare-enhanced dipole moment. Recovery to quiescent levels follows $e^{-\alpha_f \Delta t}$ with $\tau_{1/2} \approx 69$ s.

§5 Space Weather Implications

The transient F_U amplification during CMEs provides a UQFF-framework prediction for geomagnetic storm intensity: the amplification factor $F_U^{\text{pert}} / F_U^{\text{quiet}}$ correlates with Dst index magnitude, offering a novel space weather metric beyond traditional solar wind parameters.

References

- PAPER_1080: Two-Stage F_U Refinement Validator
- PAPER_418: FUSunCompleteSCmSolarCycleFinalCalibrationCalculator
- PAPER_1078: Solar Wind Flux Partitioning Budget

Key References with arXiv/DOI Identifiers

1. Abbott et al. (LIGO Scientific and Virgo Collaborations, 2016). *Observation of Gravitational Waves from a Binary Black Hole Merger.* Phys. Rev. Lett. **116**, 061102 — arXiv:1602.03837 — doi:10.1103/PhysRevLett.116.061102
2. Murphy, D. (2026). *Unified Quantum Field Framework (UQFF): Star-Magic v5.x Whitepaper Series.* Star-Magic Repository — github.com/Daniel8Murphy0007/Star-Magic